Your Name:

ID #:

Worksheet: Limits at Discontinuity - Part 2

Note: for all the problems, identify the limit as a number, $-\infty$, ∞ or DNE (does not exist)

1. Show all the work to find the following limit.

(a) Find $\lim\limits_{x\to 5} \dfrac{2x^2 - 7x - 15}{x - 5}$

Solution:

$$= \lim_{x\to 5} \frac{(2x+3)\cancel{(x-5)}}{\cancel{(x-5)}}$$
$$= \lim_{x\to 5}(2x+3)$$
$$= 2\cdot 5 + 3$$
$$= \boxed{13}$$

(b) Find $\lim\limits_{x\to 1} \dfrac{x^3 - 1}{x - 1}$

Solution:

$$= \lim_{x\to 1} \frac{(x^2+x+1)\cancel{(x-1)}}{\cancel{(x-1)}}$$
$$= \lim_{x\to 1}(x^2+x+1)$$
$$= \boxed{3}$$

(c) Find $\lim\limits_{x\to -2} \dfrac{x+2}{\sqrt{x+6} - 2}$

Solution:

$$= \lim_{x\to -2} \frac{(x+2)(\sqrt{x+6}+2)}{(\sqrt{x+6}-2)(\sqrt{x+6}-2)}$$
$$= \lim_{x\to -2} \frac{\cancel{(x+2)}(\sqrt{x+6}+2)}{\cancel{(x+6-4)}}$$
$$= 2 + 2$$
$$= \boxed{4}$$

(d) Find $\lim\limits_{x\to -1} \dfrac{\sqrt{x+10} - 3}{x + 1}$

Solution:

$$= \lim_{x\to -1} \frac{(\sqrt{x+10}-3)(\sqrt{x+10}+3)}{(x+1)(\sqrt{x+10}+3)}$$
$$= \lim_{x\to -1} \frac{\cancel{(x+10-9)}}{\cancel{(x+1)}(\sqrt{x+10}+3)}$$
$$= \frac{1}{\sqrt{9}+3}$$
$$= \boxed{\frac{1}{6}}$$

(e) Find $\lim\limits_{x\to 0^+} \dfrac{3e^x}{\ln x}$

Solution:

$$\lim_{x\to 0^+} \frac{3e^x}{\ln x} = \frac{\lim\limits_{x\to 0^+} 3e^x}{\lim\limits_{x\to 0^+} \ln x} = \frac{3}{-\infty} = 0$$

(f) Find $\lim\limits_{x\to 0} \dfrac{x^2 + 8sin(x)}{x}$

Solution:

$$= \lim_{x\to 0}\left(\frac{x^2}{x} + \frac{8sin(x)}{x}\right)$$
$$= \lim_{x\to 0} x + 8\cdot \lim_{x\to 0} \frac{sin(x)}{x}$$
$$= 0 + 8\cdot 1$$
$$= \boxed{8}$$

(g) Find $\lim_{x\to 0} \frac{tan(6x^2)+sin^2(2x)}{x^2}$

Solution:
$= \lim_{x\to 0} \frac{tan(6x^2)}{x^2} + \lim_{x\to 0} \frac{sin^2(2x)}{x^2}$
$= 6 \cdot \lim_{x\to 0} \frac{tan(6x^2)}{6x^2} + \lim_{x\to 0} (\frac{sin(2x)}{x})^2$
$= 6 \cdot 1 + \lim_{x\to 0} (2 \cdot \frac{sin(2x)}{2x})^2$
$= 6 + 2^2 \cdot 1$
$= \boxed{10}$

(h) Find $\lim_{y\to 0} (\frac{6}{y^2+y} - \frac{6}{y})$

Solution:
$= \lim_{y\to 0} (\frac{6}{y(y+1)} - \frac{6\cdot(y+1)}{y(y+1)})$
$= \lim_{y\to 0} \frac{\cancel{6} - 6y - \cancel{6}}{y(y+1)}$
$= \lim_{y\to 0} -\frac{6\cdot \cancel{y}}{\cancel{y}(y+1)}$
$= -6 \cdot \lim_{y\to 0} \frac{1}{y+1}$
$= -6 \cdot \frac{1}{0+1}$
$= \boxed{-6}$

(i) Find $\lim_{x\to 0} \frac{10sin(2x)}{1-cos(2x)}$

Solution:
$= \lim_{x\to 0} \frac{(10sin(2x))(1+cos(2x))}{(1-cos(2x))(1+cos(2x))}$
$= \lim_{x\to 0} \frac{(10sin(2x))(1+cos(2x))}{1-cos^2(2x)}$
$= \lim_{x\to 0} \frac{(10\cancel{sin(2x)})(1+cos(2x))}{sin^{\cancel{2}}(2x)}$
$= 10 \cdot \lim_{x\to 0} \frac{(10)(1+cos(2x))}{sin(2x)}$
$= \boxed{\infty}$